Nikhil Verma

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PROFESSIONAL EXPERIENCE

Amazon **Santa Clara, CA**
Software Development Engineer I *FEB 2025 - Present*

- Shipped multi-product instance support for AWS License Manager User Subscriptions as sole low-level-design author, delivering 40+ changes across Java Lambda service logic, DynamoDB persistence, and EC2 Image Builder AMI pipelines.
- Halved customer compute cost by letting Office and Visual Studio share a single EC2 instance, replacing the prior one-product-per-instance model.
- Implemented a distributed cross-account user-subscription workflow in Java: modeled the CrossAccountRelationship entity, migrated a standalone DynamoDB table into a GSI, and built sync and async authorization across Lambda, Step Functions, and DynamoDB, lowering customer billing by $10 per user per month.
- Led a security analysis of Visual Studio build components that showed a static, non-rotatable product key could be captured on-host (Windows Event ID 4688). Rejected 4 delivery designs and prevented a fleet-wide product-key leak.
- Onboarded the full User Subscriptions service (206 production accounts across 34 regions) to the AWS DevOps agentic-ops platform via CDK, integrating Ticketing, Medic, Rome, MWS, and Timber across 16 pipelines.
- Automated triage of roughly 800 tickets a month with Ticket Genie, an internal Python multi-agent tool on Amazon Bedrock and Claude, Lambda and Step Functions, cutting time to resolution by 30%.
- Extended Java backend services and REST APIs to add Windows Server 2025 support and built the backend for Global Licensing Insights, an AWS re:Invent launch. Integrated License Manager data with Tagris for tag-based reporting and added CloudWatch metrics, alarms, dashboards, and canaries for cloud readiness (ORR).

Amazon **Santa Clara, CA**
AI Engineer Intern *JUN 2024 - SEP 2024*

- Developed a multi-agent code-generation pipeline that converts complex legal documents into machine-readable licensing statements, saving 100+ manual hours a month.
- Orchestrated LangGraph, LangChain and ReAct agents over ChromaDB and OpenSearch vector stores on AWS Bedrock, writing the agent-tool layer from scratch because MCP had not yet released.
- Designed a retrieval-augmented workflow with semantic chunking and similarity search, reaching 83% exact-statement accuracy in internal tests.

DEPARTMENT OF INFORMATICS, UNIVERSITY OF CALIFORNIA, IRVINE **Irvine, CA**
Graduate Student Research Assistant *JAN 2024 - DEC 2024*

- Benchmarked an accessible ChatGPT build (Next.js, MongoDB, Auth0, OpenAI APIs) that makes slide presentations readable for blind users, on NSF-funded MLLM research under Dr. Anne Marie Piper.
- Built an object-oriented Python abstraction over Slack's API that cut self-reported frustration during query tasks by 75% for visually impaired users in study measures.

AMADEUS SOFTWARE LABS INDIA PVT LTD **Bangalore, India**
Full Stack Developer - I *AUG 2021 - AUG 2023*

- Owned the full lifecycle of an in-house data-driven automation project (Golang, Python, React) from concept to delivery, saving 3 hours of manual work a day, cutting processing time by about 70%, and fully automating bug-reporting workflows.

SKILLS

Backend & Systems: Distributed Systems, Microservices, Event-Driven Architectures, REST APIs, Observability (CloudWatch)
Languages & Frameworks: Java, Python, Golang, Spring Boot, TypeScript/JavaScript, Flask, Django
Cloud & Infrastructure: AWS (Lambda, API Gateway, DynamoDB, S3, Step Functions, EC2 & Image Builder, CloudWatch, CDK), Docker, Kubernetes, Terraform, CI/CD (Jenkins, GitLab)
Databases & Storage: DynamoDB, PostgreSQL, MongoDB, OpenSearch, ChromaDB
AI & LLMs: Multi-Agent Systems (LangGraph, LangChain, Strands), Amazon Bedrock AgentCore, Model Context Protocol (MCP), Agentic RAG, ReAct, Prompt Engineering, LLM Evaluation, Vector Databases (OpenSearch, ChromaDB), Amazon Bedrock, Claude

EDUCATION

UNIVERSITY OF CALIFORNIA, IRVINE **Irvine, California**
Master of Computer Science *SEP 2023 - DEC 2024*
CGPA: 3.91/4 • Coursework: Artificial Intelligence, Generative Models, Advanced Programming and Problem Solving, Operating Systems

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY **Patiala, India**
Electronics and Computer *JUN 2017 - JUN 2021*
CGPA: 3.68/4 • Coursework: Data Structures and Algorithms, Operating Systems, Cybersecurity

PUBLICATIONS & INVENTIONS

Invention disclosure (primary inventor): Filed with Amazon in the area of AI cost governance; built a working prototype. Details confidential pending patent review.

N. Verma, S. Kumari and P. Jain, "Post Quantum Digital Signature Change in IOTA to Reduce Latency in Internet of Vehicles (IoV) Environments," 2022 International Conference on IoT and Blockchain Technology. Improved IOTA encryption speed by 81% by replacing W-OTS with Dilithium for IoV networks.

PROJECT

XTC COIN

- Created a Python blockchain-style cryptocurrency backend: ledger API, multi-peer support, pytest suite, virtualenv tooling.